BIM-based Variant Retrieval of Building Designs Using Case-based Reasoning and Pattern Matching

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Abstract

Architects today still very often create traditional 2d plans in the early planning phase, which form the basis for discussions with investors, clients and engineers. The designs are often difficult to understand for non-experts and design variants can only be compared with great effort. The use of BIM models simplifies both communication and the actual creation through the use of parametric modeling approaches. Even though a building project always has very individual characteristics, it makes sense to take experience from other projects as a basis for the design. Therefore, it may be that suitable floor plans of similar buildings already exist, which are a good basis for variant studies in early design phases. A challenge is to be able to access the experience of other architects. Currently, this information is not available and is difficult to transfer from one project to the next. This paper outlines a solution for a BIM-based variant retrieval of a building design in early design phases using Case Based Reasoning (CBR) and Pattern Matching (PM). On the one hand, this offers the possibility to learn from old mistakes, and on the other hand, it enables the quick selection of suitable variants for a building from a diverse pool of variants.

Keywords –

Building information Modeling (BIM), case-based reasoning (CBR), Early Design Phases, Industry Foundation Classes (IFC), Pattern Matching (PM), Variant Retrieval, Similarity

1 Introduction

In the planning phase, referencing building concepts is a common practice, for example to find and evaluate similar building concepts and use them for a new project. Such manual search can take an unacceptable amount of time. An automatic generation of suitable designs or variants promises a faster workflow and at the same time serves as a transparent assessment of the building project and various analytical parameters, which is not insignificant for the consideration of the variants themselves [14]. Variant matching always allows a suitable thematic and technical view (construction costs, sustainability, etc.) due to the attributes stored in the Industry Foundation Class (IFC) interfaces, as the effects of the change in parameters and components are directly visible in the efficiency of the building. In order to find variants, the case-based reasoning method can be useful [7]. This approach starts with a problem (search query) that correlates with the specific ideas about the building project and the concepts of the planners. This paper presents a solution for finding suitable variants in the early design phases of buildings, using the case-based reasoning (CBR) approach based on the IFC standard. For this purpose, specific case studies are provided to explain the similarity calculation used, which is essential for the first phase of the CBR cycle. Building on the basic idea of CBR (similarity recognition) the method of pattern matching (graphical pattern recognition) is used to illustrate the selection of suitable variants with given framework conditions (search queries).

In previous research, a concept for modelling and managing design options was already created, which is connected to the BIM models. Three variant classes (structural, functional and product variants) were defined and implemented using the IFC [11]. This procedure avoids redundancy and offers a wide range of applications. Furthermore, a representation of variants in using graph theory was introduced [10].

The basic idea of finding suitable variants of floor plans already appeared in 1996 in a journal article by Gomez de Silva Garza and Maher [9]. At that time the technical possibilities regarding BIM models did not exist. Langenhan et al. [13], address this issue and attempt to match building floor plans using the fingerprint method and a database matching. BIM is not integrated here either. Similar to us, Althoff et al. [14] use the CBR approach with the help of exact and inexact graph matching to support the architect in finding similar 2d floor plans in the early design phases. The properties and consequences of components and structures cannot be identified. In the reviewed literature, there is no

基于BIM的建筑设计变体检索：采用案例推理与模式匹配方法

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摘要

如今，建筑师在规划的早期阶段仍经常绘制传统的二维平面图，这些图纸构成了与投资者、客户及工程师进行讨论的基础。然而，对于非专业人士而言，这些设计往往难以理解，且设计方案之间的比较也极为费力。采用BIM模型借助参数化建模方法，不仅能够简化沟通流程，还能显著提升设计效率。尽管每个建筑项目都具有高度的个性化特征，但在设计过程中借鉴其他项目的经验仍具有重要意义。因此，很可能已经存在适用于类似建筑的标准化平面布局，这些布局可作为早期设计阶段开展方案比选的良好基础。然而，如何有效获取并利用其他建筑师的设计经验仍是一项挑战；目前，此类信息尚无法便捷获取，且难以在不同项目之间实现有效传递。

本文提出了一种基于BIM的建筑方案检索解决方案，该方案在设计早期阶段采用基于案例的推理（CBR）与模式匹配（PM）技术。一方面，这有助于从以往的设计失误中汲取经验教训；另一方面，它能够从丰富的方案库中快速筛选出适用于特定建筑项目的合适方案。

关键词 –

建筑信息建模（BIM）、基于案例的推理（CBR）、早期设计阶段、工业基础类（IFC）、模式匹配（PM）、变体检索、相似性

1 引言

在规划阶段，参考既有建筑方案是一种常见做法，例如用于查找并评估相似的建筑方案，并将其应用于新项目。然而，这种手动检索往往耗时过长，难以接受。而自动生成合适的方案或变体，则有望加快工作流程，同时为建筑项目及其各项分析参数提供透明的评估依据，但这并不对于各方案本身的考量而言，其影响可忽略不计[14]。由于存储在工业基础类（IFC）接口中的属性，方案匹配能够始终提供恰当的功能性与技术性视角（如建造成本、可持续性等），并且参数与构件的变化对建筑效能的影响可直接显现。为寻找可行的方案，基于案例的推理方法颇具价值[7]。该方法以一个问题（检索查询）为起点，该问题与建筑设计的具体构想及规划者的理念相呼应。本文提出了一种解决方案，旨在利用基于IFC标准的基于案例的推理（CBR）方法，在建筑的早期设计阶段筛选出合适的方案。为此，本文提供了若干具体案例，用以阐释所采用的相似度计算方法，而该方法是CBR循环第一阶段的核心环节。在CBR基本思想（相似性识别）的基础上，进一步采用模式匹配方法（图形模式识别），以说明在给定约束条件（检索查询）下如何遴选适宜的方案。

在先前的研究中，已提出了一种用于建模与管理设计方案的构想，并将其与BIM模型相衔接。研究定义并基于IFC标准实现了三种变体类别（结构变体、功能变体和产品变体）[11]。该方法可避免冗余，并具有广泛的应用前景。此外，还引入了基于图论的变体表达方式[10]。

寻找合适平面布局变体的基本思路早在1996年就已见诸Gomez de Silva Garza与Maher发表的一篇期刊论文[9]。当时，关于BIM模型的技术条件尚不具备。Langenhan等人[13]也关注了这一问题，并尝试采用指纹法及数据库匹配来实现建筑平面图的比对，但同样未将BIM技术纳入其中。与我们类似，Althoff等人[14]借助精确与非精确的图匹配方法，基于案例推理（CBR）框架，在设计早期阶段支持建筑师检索相似的二维平面布局，然而无法识别构件与结构的属性及其影响。在现有文献中，尚未发现

evidence of the correlation between BIM models and variant decision-making.

This paper is dealing with the topic in order to offer a different and evolved solution that integrates BIM. This paper addresses the issue to provide an evolved solution that integrates BIM models and uses IFC for the implementation. These factors are essential for interdisciplinary collaboration during the early design phases of buildings and are neglected by the authors mentioned above.

2 Background and methodology

The following chapters briefly summarize the basic theoretical knowledge and methodology on which the paper is based on.

2.1 BIM in context of the research topic

The acronym BIM stands for Building Information Modeling, which has been increasingly influencing the construction industry for several years. In a digital building model, the exact geometry of the building and information on all the built-in components are stored within the model and recorded over the entire life cycle of the building. Thus, this information can be used for (facility) management, revitalization, expansion or demolition. Additionally, a loss-free exchange of files with investors, energy and construction engineers or facility management is possible over the entire life cycle of the building [3].

The exchange of this (open) BIM data is made possible by the Industry Foundation Classes (IFC), an international open standard (ISO 16739) developed by buildingSMART [6]. The IFC is continuously being further developed since 2013 and is available in the latest version of the IFC standard as IFC4 [12].

BIM is compatible with our research work because the IFC interface can be used to define the information range of individual elements, directly show the changes of parameters and thus allow a comparison of parameters among themselves and thematically, regarding the whole design. Therefore, structure, function and product variants can be assigned to the respective level of detail (LOD). Three variant types have already been defined [11]. The structure variant offers options to the structure of the building, for example the geometry of the building or the number of building storeys. The functional variant can be exemplified by the load-bearing structure of the building or objects that fulfil the same use or purpose. The product variant, as the smallest level of the variant types, includes individual objects (e.g. doors) that can be exchanged with similar objects or other property values, without affecting the structure and function.

Using BIM and IFC makes it possible to capture different variants with the entire geometry and all properties. This means that they are available to everyone as explicit knowledge. For the research in this topic, a graph-based representation of the data structure is chosen [10]. In this, the BIM model is represented based on the IFC standard with the different entities, attributes and, consequently, the geometry. This continues the previous considerations in the research context and is suitable for the methodological approach of CBR and pattern match (PM). For this purpose, patterns must be defined, which in turn are linked to the similarity definitions and afterwards they can be shown as several options in a graph database.

2.2 Current problems

Currently, individual BIM models are being Currently, individual BIM models are being developed from scratch for each construction project. In this process, the planners are guided by the wishes of the client. The planner therefore designs a great many different floor plan variants for a wide variety of building types and uses during his professional career. This experience is used when a planner encounters similar boundary conditions and construction requirements. However, since this procedure is related to the implicit knowledge of the planner, the number of variants is limited. With the presented solution the knowledge of each planner is available by storing different variants of building designs in a database and generating automated solutions.

2.3 Methodology and process

2.3.1 CBR

CBR is a methodical way of problem solving, using problems that have already been solved and proven to be successful. In the context of this research, it is useful to adapt a suitable solution from a case base, as a 1:1 transfer of the solution turns out to be difficult, because buildings are mostly heterogeneous. Thus there are almost never exact matches, since the external or internal circumstances are not the same even though the appearance may be the same. That is why it is necessary to determine a degree of similarity. The usefulness of a solution is consequently an optimization phase that is based on the similarity determination of the components and adapted to their specific weighting [7].

The process of case-based reasoning can be represented as a cycle, which Aamodt and Plaza [1] divide into four steps: retrieve, reuse, revise and retain. In this paper the retrieval process is dealt with and the problem is defined as finding a suitable variant. The procedure is demonstrated by own case studies and differs from MetisCBR [14] in that it includes both, a graph structure and geometry to make the search results more accurate.

BIM模型与方案决策之间相关性的证据。

本文旨在探讨该课题，以提出一种集成BIM的差异化且更为完善的解决方案。具体而言，本文通过整合BIM模型并采用IFC标准进行实施，来解决相关问题。这些要素对于建筑早期设计阶段的跨学科协同至关重要，但却被上述作者所忽视。

2 背景与方法论

以下各章简要概述了本文所依据的基本理论知识与方法论。

2.1 研究主题背景下的BIM

缩写词BIM代表建筑信息模型，近年来其对建筑行业的影响日益显著。在数字化建筑模型中，建筑的精确几何形体以及所有内置构件的相关信息均被存储于模型内，并贯穿建筑的整个生命周期。因此，这些信息可用于设施管理、改造、扩建或拆除等环节。此外，在建筑的整个生命周期内，还可实现与业主、能源工程师、结构工程师及设施管理方之间的无损文件交换[3]。

此类（开放）BIM数据的交换得以实现，得益于由buildingSMART制定的国际开放标准——工业基础类（IFC）[6]（ISO 16739）。自2013年以来，IFC标准不断得到完善，并已在最新版本的IFC标准中以IFC 4的形式发布[12]。

BIM 与我们的研究工作具有良好的兼容性，因为 IFC 接口可用于界定单个构件的信息范围，直接展示参数的变化，从而实现各参数之间以及按主题进行的对比，并覆盖整个设计过程。因此，结构、功能和产品变体均可对应于相应的细节层次（LOD）。目前已定义了三种变体类型[11]：结构变体提供了针对建筑结构的多种选项，例如建筑的几何形态或楼层数量；功能变体则以建筑的承重结构或承担相同用途或功能的构件为例；而作为变体类型中最小层级的产品变体，则包含可与其他同类构件或具有不同属性值的构件互换的单个构件（如门），且不会影响建筑的结构与功能。

利用BIM和IFC技术，可以对包含全部几何信息的不同方案进行记录。属性。这意味着这些属性作为显性知识对所有人开放。针对该课题的研究，选择了基于图的数据结构表示方法[10]。在此方法中，BIM模型依据IFC标准进行表达，包含不同的实体、属性以及相应的几何信息。这一做法延续了研究背景中的既有考虑，并且适用于案例推理（CBR）与模式匹配（PM）的方法论框架。为此，必须定义相应的模式，这些模式又与相似性度量规则相衔接，随后可在图数据库中以多种备选方案的形式呈现。

2.2 当前问题

目前，针对每个建筑项目，BIM模型均从零开始单独构建。在此过程中，

规划师的设计工作以客户的需求为导向。在其职业生涯中，规划师会针对多种建筑类型和用途设计大量不同的平面布局方案。当遇到类似的边界条件与建造要求时，这些经验便会被加以运用。然而，由于这一过程主要依赖于规划师的隐性知识，因此所能生成的方案数量有限。而通过所提出的解决方案，将各类建筑设计方案存储于数据库中并自动生成相应方案，便可实现对每位规划师知识的有效积累与共享。

2.3 方法论与流程

2.3.1 CBR

案例推理是一种系统化的问题解决方法，其核心在于利用那些已被解决且被证明有效的先前案例。在本研究的背景下，从案例库中适配一个合适的解决方案具有重要意义，因为将该解决方案进行一对一的直接迁移往往存在困难，这是因为建筑物大多具有异质性。因此，即使外观相似，外部或内部条件也几乎不可能完全一致，从而导致几乎不存在完全匹配的案例。正因如此，有必要确定一种相似度量。相应地，解决方案的有效性评估便成为一项基于各组件相似度量化并结合其特定权重进行调整的优化过程[7]。

基于案例的推理过程可以被表述为一个循环，Aamodt 和 Plaza [1] 将其划分为四个步骤：检索、复用、修正和保留。在本文中，重点探讨了检索过程，并将问题定义为寻找合适的变体。该方法通过作者自身的案例研究加以说明，并与 MetisCBR [14] 相比，其特点在于同时考虑图结构与几何信息，以提高搜索结果的准确性。

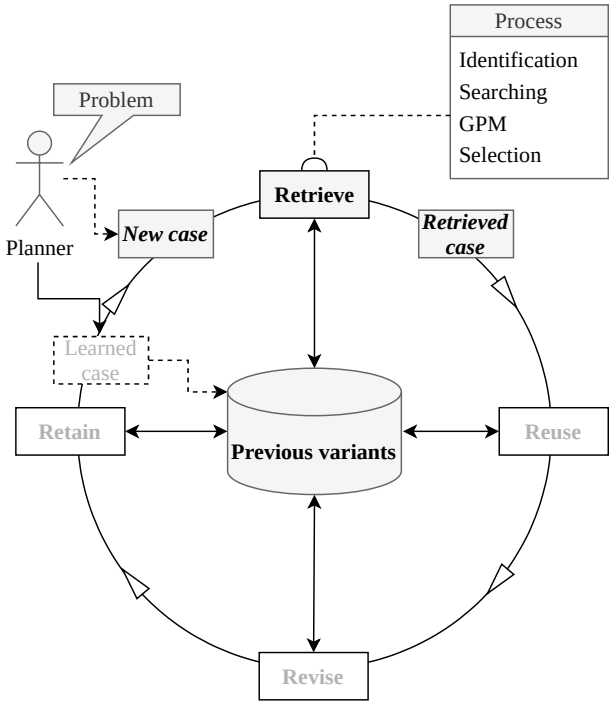


Figure 1. CBR cycle according to Aamodt & Plaza [14]

2.3.2 GPM

The method of graph pattern matching (GPM) is a possibility to find (suitable) variants from a graph database. Based on a pattern query, matching graphs are selected from a case base that provide an answer to the existing request (problem). This approach can also be extended to subgraphs if more specific requirements are being searched for. Similarity definition and weights are of great importance for the pattern matching in order to limit the results and have them target-oriented [15]. Furthermore, a categorization is made between two types of pattern matching (PM), which are called exact PM and inexact PM [14].

An exact PM is characterized by one-to-one isomorphism, i.e. the same number of nodes are contained in both graphs and both graphs have the same manner of connections (number of edges). If the graphs are not isomorph, subgraphs can be examined for isomorphism in the next step [14].

The inexact PM is a different approach that is more suitable to the heterogeneous field of the building sector, because building structures and room uses differ from each other. The search query is decomposed into different IFC data or building components and these are evaluated and normalized with a similarity score [14].

In addition, specific weights are added, which allows the similarities to be determined and the relevance for the comparison of variants to be specified. A variant is a match if the graph of the variant is a subgraph of the database entry. In contrast to the exact pattern match, so-called replacement rules are applicable to the inexact PM [14]. These replacement rules are important, because they extend the inexact pattern match similarity. For example, instead of a load-bearing wall, the user is able to choose a load-bearing column. or instead of a connection between two rooms with a door, a breakthrough is possible. These replacement rules must be individually adapted by the user.

2.3.3 Similarity

In the following, the similarity calculation according to Richard Hemming is determined, which is often used in the retrieval process of the CBR. The concept of generalized similarity determines the similarity of two cases for any number of attribute values, which can be weighted and normalized (1). According to Hemming, a qualitative and quantitative similarity determination is possible. Generalized similarity allows a comparison of normalized real attributes, as well as a comparison based on classifications [2].

$$\text{sim}\,(p,v) = \frac{\sum_{i=1}^{n} \omega_i sim_i(p_i, v_i)}{\sum_{i=1}^{n} \omega_i} \quad (1)$$

Using this formula, the similarity between a problem *p* and a possible variant *v*, with non-negative weights, is a value between zero and one. The more similar the variants, the higher the value of sim [12]. According to this approach, the similarities are used for all entities stored in the IFC data structure. Depending on the type of entities, similarity measures can be defined for both IFC quantities and property definitions.

2.3.4 Retrieval

In a database (case base), *n* variant graphs are defined, which differ from the simple consideration of the building graphs in that they contain option points. These variant graphs were defined as part of other projects and include structural, functional and product variants. Following the top-down design, the new user specifies the various parameters for the current project (problem) for which he wants to find a variant (e.g. use of the building floor, sizes, number and functions of the rooms). The process also allows searching for required connections (e.g. between the rooms). It is up to the user to decide whether this is necessary for the problem or not.

After that, the individual entities are weighted by the user, which results in priorities being selected depending on individual preferences and the use cases, generating different solutions in the end. The database is searched for graphs that are either an exact match or an inexact match of the graph of the problem. These matches are presented to the user and sorted by the magnitude of the

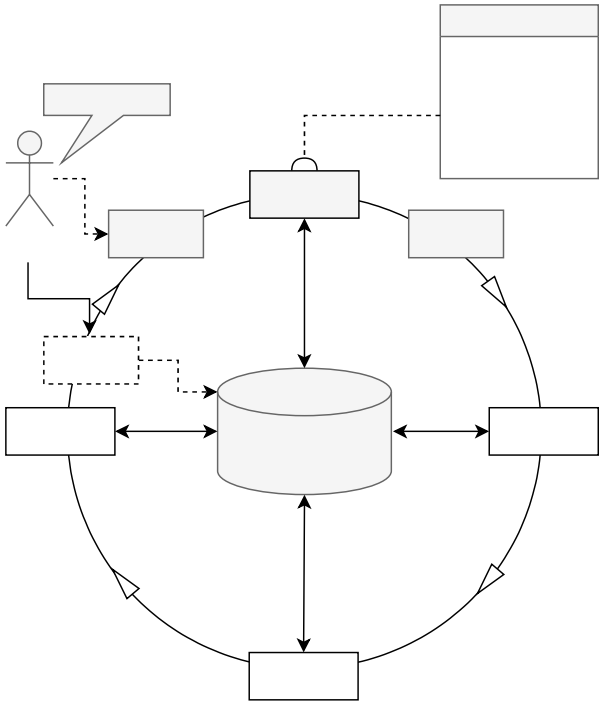


图1. 根据Aamodt与Plaza [14]提出的案例推理循环

2.3.2 GPM

图模式匹配（GPM）方法为从图数据库中寻找（合适的）变体提供了一种可能性。基于模式查询，系统会从案例库中筛选出与之匹配的图，从而对当前请求（问题）给出解答。当需要满足更为具体的要求时，该方法还可扩展至子图层面。在模式匹配过程中，相似性定义与权重的设定至关重要，它们有助于限定结果范围并使其更具针对性[15]。此外，模式匹配还被划分为两种类型，即精确模式匹配与非精确模式匹配[14]。

一个精确的PM模型以一对一同构为特征，即两个图包含相同数量的节点，并且两者的连接方式（边的数量）也完全一致。如果两个图不满足同构条件，则可在下一步对子图进行同构性检验[14]。

不精确匹配法是一种更为适合建筑行业这一异质性领域的不同方法，因为建筑结构和房间用途各不相同。搜索查询被分解为不同的IFC数据或建筑构件，并通过相似度评分对其进行评估和归一化处理[14]。

此外，还引入了特定的权重，从而能够确定相似性，并明确各变体比较的相关性。当某一变体的图是另一图的子图时，该变体即被视为匹配。

数据库条目。与精确模式匹配不同，所谓的替换规则适用于不精确的部件模型匹配[14]。这些替换规则至关重要，因为它们能够扩展不精确模式匹配的相似性范围。例如，用户可以选择承重柱来代替承重墙，或者选择开洞来代替两间房间之间的门连接。这些替换规则必须由用户进行个性化配置。

2.3.3 相似性

在下文中，将依据理查德·海明的方法确定相似度计算，该方法常用于案例推理（CBR）的检索过程。广义相似度的概念可用于确定任意数量属性值之间的相似性，这些属性值可进行加权和归一化处理（1）。根据海明的观点，可以实现定性和定量的相似度判定。广义相似度既允许对归一化的实值属性进行比较，也支持基于分类的比较[2]。

$$\mathrm{sim}\,(p,v) = \frac{\sum_{i=1}^{n} \omega_i sim_i(p_i, v_i)}{\sum_{i=1}^{n} \omega_i} \qquad (1)$$

采用该公式，对于具有非负权重的问题p与可能的变体v之间的相似度，其取值范围介于0至1之间。变体之间的相似度越高，sim的值就越大[12]。根据这一方法，相似度计算被应用于存储在IFC数据结构中的所有实体。根据实体的类型，既可以针对IFC量定义相似度度量，也可以针对属性定义相似度度量。

2.3.4 检索

在数据库（案例库）中，定义了n个变体图，这些变体图与单纯考虑建筑图有所不同，因为它们包含可选节点。这些变体图是在其他项目中定义的，涵盖了结构、功能和产品层面的变体。按照自顶向下的设计方法，新用户为当前项目（问题）指定各项参数，以寻找相应的变体方案（例如建筑楼层的使用方式、各房间的尺寸、数量及功能）。该流程还支持对所需连接关系的检索（例如房间之间的连接）。是否需要进行此类检索由用户根据具体问题自行决定。

此后，各实体项由用户进行加权，从而根据用户的个人偏好与具体用例确定优先级，最终生成不同的解决方案。系统会在数据库中检索与问题图结构完全匹配或部分匹配的图谱，并将这些匹配结果呈现给用户，同时按其匹配程度的大小进行排序。

calculated similarity. If the result still has a need for optimization, fine-tuning can be done. This is helpful to show possible further structure, function or product variants for a selected area if a match is identified. This might be the case if a room contains windows or doors as a result, which in turn can be replaced by other product variants. Another case could be chosen other wall structures, for example, to improve the energy properties of the building (structure variant) (Fig. 2).

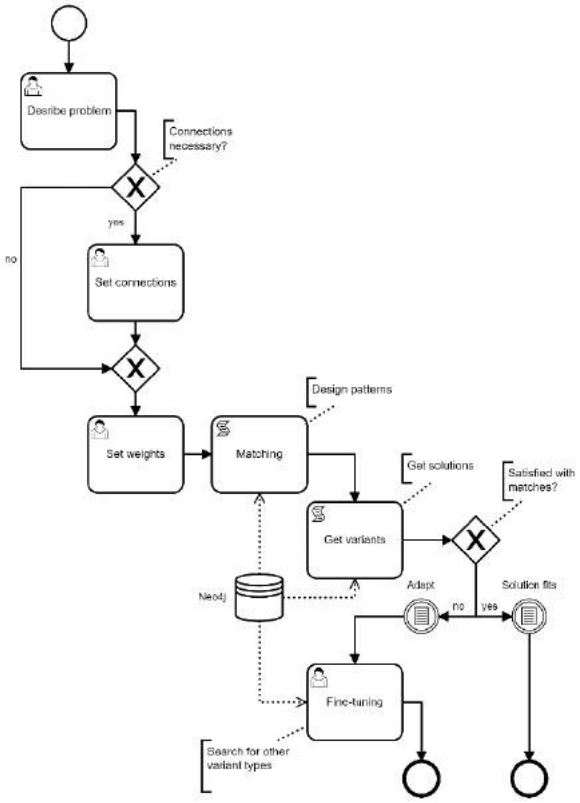


Figure 2. BPMN (Retrieve)

3 Case study

With the help of a simplified case study, the BIM-based variant retrieval is presented. In determining the weights, planners and architects usually act subjectively, stick to their experience, are oriented to the sustainability or adapt the wishes of the clients [4]. In this paper the authors assume the role of the user, which is why the problem description and weights are determined by them.

A variant is to be found for a ground floor with office use, an area of 290 m^2 and with at least two offices. Other criteria set by the user are, offices with windows, a connection of the offices with a corridor, the wall thicknesses and other attributes. According to this problem description, the floor plan could possibly look like the following which captures the problem visually, but is not part of the data base (Fig. 3).

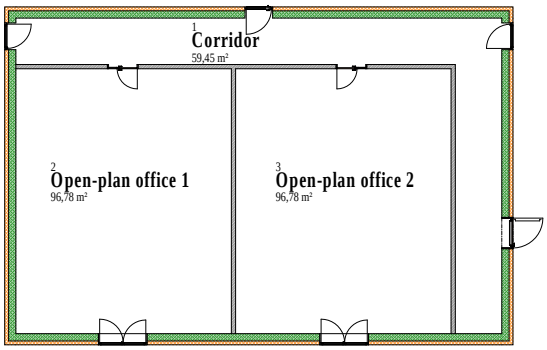


Figure 3. Exemplary floor plan according to problem description of the use

The comparison of the problem with different variants from a case base is based on the different attributes, stored in the BIM model. Table 1 lists the attributes essential for the case study, together with their defined specifications. Based on these attributes, the most similar problems in the current case base are determined for the defined issue.

Table 1. Attributes and specifications

Entities	IfcObject-Quantities	Property definitions
IfcBuilding-Storey	GrossFloorArea GrossVolume	-
IfcColumn	AverageHeight GrossFloorArea GrossVolume	LoadBearing
IfcDoor	OverallHeight OverallWidth	IsExternal FireBearing
IfcRelConnects	-	-
IfcSpace	GrossFloorArea GrossVolume	Category
IfcWall IfcWindow	NormalWidth OverallHeight OverallWidth	LoadBearing IsExternal

The calculation of the similarity measure must be defined for each property. For this purpose, the individual properties are mapped to integers in order to be able to calculate normalized differences. The properties are given values to make strings and deviations measurable and to be able to measure the differences. At the end of the table there is a normalization factor (NF) to normalize the differences and receive a number between zero and one. A difference of zero corresponds to the similarity measure one, that means, that the attributes are equal to each other. A difference of one equates to the similarity measure zero, that means that the properties are not similar to each other.

计算得到的相似度。如果结果仍需优化，可进一步进行微调。这有助于在识别出匹配项时，为选定区域展示可能的进一步结构、功能或产品变体。例如，当房间中包含窗户或门时，这些构件又可被其他产品变体所替代。另一种情况是选用不同的墙体构造方案，以提升建筑的能源性能（构造变体）（图2）。

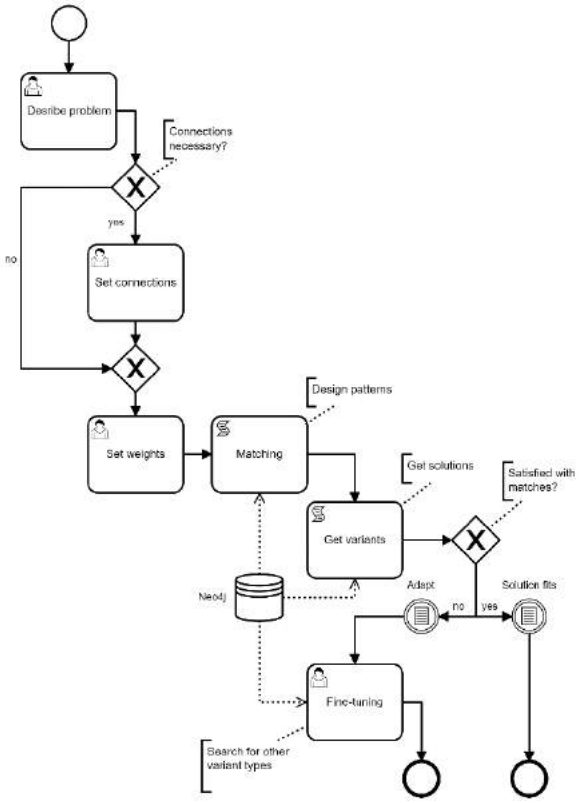


图2. BPMN（检索）

3 案例研究

借助一个简化的案例研究，本文展示了基于BIM的方案检索方法。在确定权重时，规划师和建筑师通常凭主观判断，依赖自身经验，或以可持续性为导向，亦或迎合客户的需求[4]。在本文中，作者扮演用户角色，因此问题描述及权重均由其自行确定。

针对一层办公用途的建筑，存在一种可行的方案，其建筑面积为290平方米，并至少包含两个办公室。用户设定的其他条件包括：办公室应设有窗户、各办公室需通过走廊相连、墙体厚度及其他相关属性。根据这一问题描述，该层的平面布局可能呈现如下形式：例如下图，该图以可视化方式呈现了问题，但并不属于数据库的一部分（图3）。

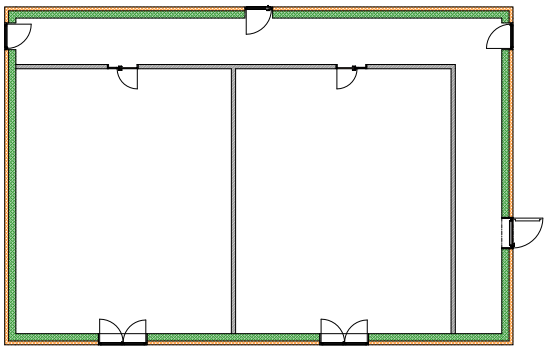


图3. 根据使用问题描述的示例平面图

基于存储在BIM模型中的不同属性，对问题与案例库中不同变体进行比较。表1列出了本案例研究所需的关键属性及其定义的规格。根据这些属性，针对所界定的问题，确定当前案例库中最相似的问题。

表1. 属性与规格

Entities	IfcObject-Quantities	Property definitions
IfcBuilding-Storey	GrossFloorArea GrossVolume	-
IfcColumn	AverageHeight GrossFloorArea GrossVolume	LoadBearing
IfcDoor	OverallHeight OverallWidth	IsExternal FireBearing
IfcRelConnects	-	-
IfcSpace	GrossFloorArea GrossVolume	Category
IfcWall	NormalWidth	LoadBearing
IfcWindow	OverallHeight OverallWidth	IsExternal

相似性度量的计算必须针对每种属性分别定义。为此，将各属性映射为整数，以便计算归一化的差异。通过为属性赋予具体数值，使字符串及其偏差可量化，并能够衡量它们之间的差异。在表格末尾设有一个归一化因子（NF），用于对差异进行归一化处理，从而得到一个介于0与1之间的数值。当差异为0时，对应的相似性度量为1，即表明这些属性彼此完全相等；当差异为1时，对应的相似性度量为0，即表明这些属性彼此不相似。

Table 2-6. Values for similarity determination

Boolean	Value
True	0
False	1

Deviation area & volume	Value
0%	0
< 5%	1
5% up to < 10%	2
10% up to < 15%	3
15% up to < 20%	4
20% and above	5
Normalization factor (NF)	1/6

Deviation height & width	Value
0cm	0
< 5cm	1
5cm up to < 10cm	2
10cm up to < 20cm	3
20cm up to < 30cm	4
30cm and above	5
NF	1/6

Deviation NominalWidth	Value
0cm	0
< 1cm	1
1cm up to < 2cm	2
2cm up to < 3cm	3
3cm up to < 5cm	4
5cm up to < 7cm	5
7cm up to < 10cm	6
10cm up to < 20cm	7
20cm and above	8
NF	1/9

Category (DIN 277)	Value
NUF 1	0
NUF 2	1
NUF 3	2
NUF 4	3
NUF 5	4
NUF 6	5
NF	1/6

Weights must be added to the selected properties to capture the relevance of the properties and solve the problem. The list of weights and the calculation of the corresponding values are subjective estimations of the designer. These weights are selected in this example as follows (Tab. 7).

Table 7. Entity weighting

IfcData	Weights
IfcBuildingStorey	1,0
IfcColumn	0,2
IfcDoor	0,3
IfcRelConnects	0,5
IfcSpace	0,6
IfcWall	0,7
IfcWindow	0,3

After these steps, the similarity measure between the problem and two possible variants can be calculated. The example mentioned at the beginning is the problem (p) for which a solution is to be found that corresponds to the weights (ω) with the specific attributes (a) of the variants (v). The values of the attributes are calculated according to the following formula [2]:

$$\begin{aligned} \mathrm{sim}(p(a_1), v(a_1)) &= \omega_1(1 \\ &\quad - (NF_1(p(a_1) - v\,(a_1)))) \\ \mathrm{sim}(p(a_1), v_1(a_1)) &= 1(1 - (\frac{1}{6}(0-0))) \\ \mathrm{sim}(p(a_1), v_1(a_1)) &= 1 \end{aligned} \quad (2)$$

In this example (2), the similarity of the IfcBuildingStorey between the problem and variant one is shown. The calculation is made for each quantity and property of the entities (Tab. 1).

The similarities of the entities are calculated, with regards to the problem, and compared in the following table (Tab. 8), with the characteristics of the individual attributes. In this process, the inputs requested by the user are compared on the basis of their IFC data with already solved problems from the case base.

Table 8. Similarity

Entities	Variant 1	Variant 2
IfcBuildingStorey	1	1
IfcColumn	0,2	0
IfcDoor	0,1	0,3
IfcRelConnects	0,5	0,4
IfcSpace	0,3	0,6
IfcWall	0,25	0,5
IfcWindow	0,15	0,3

The floor plans of the two variants are compared pictorially in order to show the differences to the problem

表2-6. 相似性判定的取值

Boolean	Value
True	0
False	1

Deviation area & volume	Value
0%	0
< 5%	1
5% up to < 10%	2
10% up to < 15%	3
15% up to < 20%	4
20% and above	5
Normalization factor (NF)	1/6

Deviation height & width	Value
0cm	0
< 5cm	1
5cm up to < 10cm	2
10cm up to < 20cm	3
20cm up to < 30cm	4
30cm and above	5
NF	1/6

Deviation NominalWidth	Value
0cm	0
< 1cm	1
1cm up to < 2cm	2
2cm up to < 3cm	3
3cm up to < 5cm	4
5cm up to < 7cm	5
7cm up to < 10cm	6
10cm up to < 20cm	7
20cm and above	8
NF	1/9

Category (DIN 277)	Value
NUF 1	0
NUF 2	1
NUF 3	2
NUF 4	3
NUF 5	4
NUF 6	5
NF	1/6

必须为所选属性赋予权重，以反映各属性的重要性并解决问题。权重的确定及其对应数值的计算均基于设计者的主观估计。在本例中，这些权重的取值如下（表7）。

表7. 实体权重

IfcData	Weights
IfcBuildingStorey	1,0
IfcColumn	0,2
IfcDoor	0,3
IfcRelConnects	0,5
IfcSpace	0,6
IfcWall	0,7
IfcWindow	0,3

在完成上述步骤后，即可计算该问题与两种可能备选方案之间的相似度。开篇所提及的示例即为待求解的问题（p），其目标是找到与备选方案（v）的特定属性（a）及相应权重（ω）相匹配的解决方案。各属性的取值按照如下公式计算[2]:

$$\begin{aligned} \mathrm{sim}(p(a_1), v(a_1)) &= \omega_1(1 \\ &\quad - (NF_1(p(a_1) - v\,(a_1)))) \\ \mathrm{sim}(p(a_1), v_1(a_1)) &= 1(1 - (\frac{1}{6}(0-0))) \\ \mathrm{sim}(p(a_1), v_1(a_1)) &= 1 \end{aligned} \quad (2)$$

在本示例（2）中，展示了问题与变体一之间IfcBuildingStorey的相似度。计算针对各实体的数量和属性分别进行（表1）。

针对该问题，计算各实体之间的相似度，并将其与各个属性的特征一并在下表（表8）中进行对比。在此过程中，系统依据IFC数据，将用户输入的条件与案例库中已解决的问题进行比对。

表8. 相似性

Entities	Variant 1	Variant 2
IfcBuildingStorey	1	1
IfcColumn	0,2	0
IfcDoor	0,1	0,3
IfcRelConnects	0,5	0,4
IfcSpace	0,3	0,6
IfcWall	0,25	0,5
IfcWindow	0,15	0,3

为便于展示与问题相关的差异，对两种方案的平面布置图进行了直观对比。

(Fig. 4 and 5).

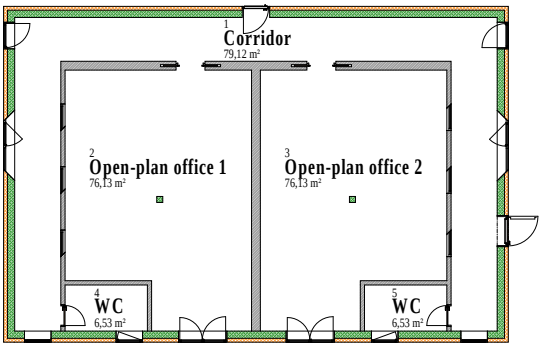


Figure 4. Building floor plan representation variant one

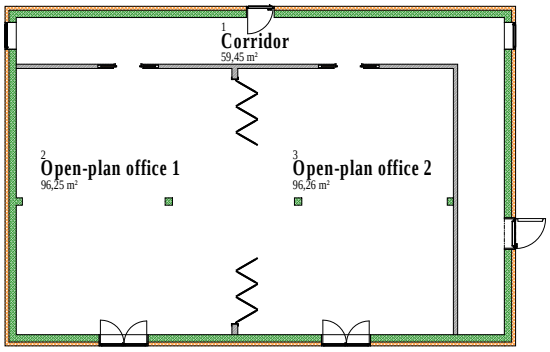


Figure 5. Building floor plan representation variant two

Compared to figure 3, the two possible solutions have differences that are more or less significant.

However, instead of the image representation, the actual process matches the IFC data structure of different projects from the case base with the existing problem. The graph of the variants one and two are inexact pattern matches out of the IFC structure of the problem definition. The representation of the floor plans is thus only representative in order to have a better look at the differences, which is more difficult to see in the graphical representation.

In figure 6, a simplified representation has been chosen to show the graphical structure of the problem. The individual properties of the data are stored in the nodes and edges.

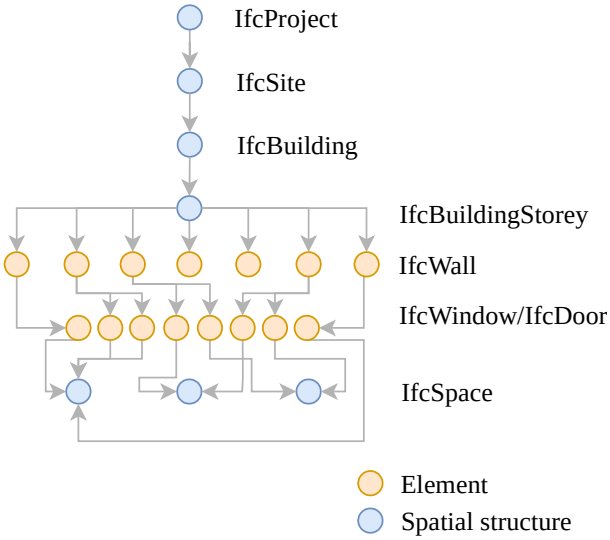


Figure 6. IFC structure of the problem definition

According to the similarity calculation, a similarity of variant one (x) to the problem (p) is calculated as follows:

$$\begin{aligned} &\mathrm{sim}(p,x) \\ &= \frac{1 + 0{,}2 + 0{,}1 + 0{,}5 + 0{,}3 + 0{,}25 + {,}015}{1 + 0{,}2 + 0{,}3 + 0{,}5 + 0{,}6 + 0{,}7 + 0{,}3} \\ &\mathrm{sim}(p,x) = 0{,}694 \end{aligned} \tag{3}$$

For variant two (y), the similarity is calculated in the same way.

$$\begin{aligned} &\mathrm{sim}(p,y) = \frac{1 + 0 + 0{,}3 + 0{,}4 + 0{,}6 + 0{,}5 + 0{,}3}{1 + 0{,}2 + 0{,}3 + 0{,}5 + 0{,}6 + 0{,}7 + 0{,}3} \\ &\mathrm{sim}(p,x) = 0{,}86 \end{aligned} \tag{4}$$

In this case, it indicates that variant two has a greater similarity to the problem and thus results as the more suitable solution regarding this issue.

If the user is not completely satisfied with this solution, he or she can search for possible alternatives to the variant types in the fine-tuning section. An example of a procedure in this area is illustrated in figure 7.

(Fig. 4 and 5).

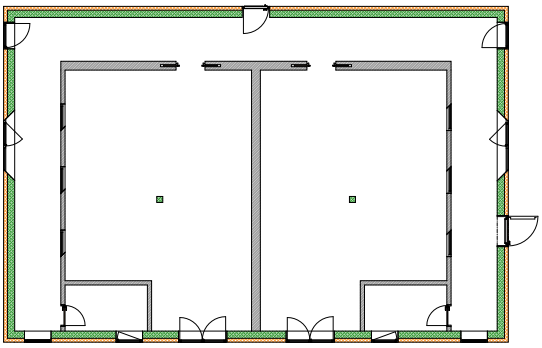


图4. 建筑楼层平面图表示方案一

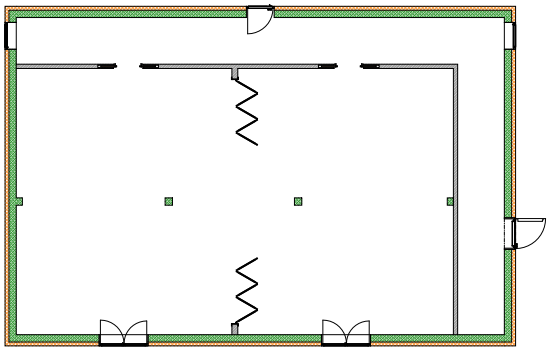


图5. 建筑楼层平面图表示方案二

与图3相比，两种可能的解决方案在不同程度上存在显著差异。

然而，与基于图像的表示方法不同，该过程是将案例库中不同项目的IFC数据结构与当前问题进行匹配。对于变体一和变体二，其图结构与问题定义的IFC结构之间存在不精确的模式匹配。

因此，平面图的表达仅具有示意性，以便更清晰地展示在图形表示中较难辨识的差异。

在图6中，采用了一种简化的表示方法来展示该问题的图结构。数据的各项属性分别存储于节点和边中。

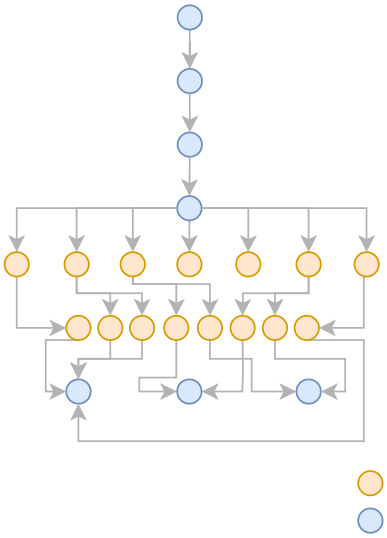


图6. 问题定义的IFC结构

根据相似度计算，变体一（x）与问题（p）的相似度计算如下：

$$\text{sim}(p,x) = \frac{1+0{,}2+0{,}1+0{,}5+0{,}3+0{,}25+{,}015}{1+0{,}2+0{,}3+0{,}5+0{,}6+0{,}7+0{,}3} \quad (3)$$

$$\text{sim}(p,x) = 0{,}694$$

对于变体二（y），相似度的计算方式与前述相同。

$$\text{sim}(p,y) = \frac{1+0+0{,}3+0{,}4+0{,}6+0{,}5+0{,}3}{1+0{,}2+0{,}3+0{,}5+0{,}6+0{,}7+0{,}3} \quad (4)$$

$$\text{sim}(p,x) = 0{,}86$$

在这种情况下，这表明变体二与该问题的相似度更高，因此被视为针对该问题更为合适的解决方案。

如果用户对该解决方案仍不完全满意，可在微调模块中查找该变体类型的其他可能替代方案。该领域的操作示例如图7所示。

```
MATCH (c:IfcElement)<-->(d:IfcElement)<--(e:IfcSpatialStructureElement), (d:IfcElement)<--(f:IfcSpatialStructureElement)
WHERE c.Entity="IfcOpeningElement" AND 0.19<=c.Depth<=0.21 AND 0.8<=c.Height<=0.9 AND 1.8<=c.Width<=2.3
    AND d.Entity="IfcDoor" AND d.`Operation Type`="SINGLE_SWING_RIGHT" AND d.IsExternal=false
    AND e.Entity="IfcSpace" AND e.`Room key Usage type`="1 Living"
    AND f.Entity="IfcSpace" AND f.`Room key Usage type`="2 Office work"
    AND size( (d)-[:IfcRelSpaceBoundary]-() ) >= 2
RETURN c.d.e.f.ID(d)
```

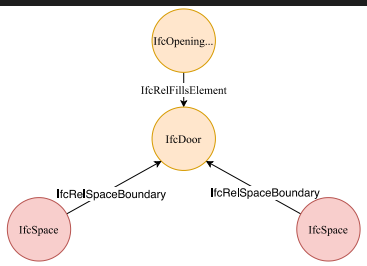


Figure 7. Optional product variant (PM)

In the given case the user is able to find an optional product variant (door) out of Neo4j based on specific deviations. By using the given cypher query, the database is searched for a door that fits into a given wall opening (depth, height, width). The object is an interior swinging door that opens to the right and connects an office use with a living use. Exact or inexact pattern matches are searched in the database that match the selected variant for which there is an option for the variant type.

Matching alternative variants are returned and the user can choose a suitable option or expand the search. This process can be adapted for each product variant and extended to functional and structural variants if required.

4 Conclusion and future work

With the help of CBR, existing knowledge can be used to find solutions for similar problems. The similarity measures can be flexibly adapted to the individual requirements of the user.

In this paper, a BIM-based solution for the retrieval process in the early design phase of a building is identified. This makes it possible to include all the information stored in a BIM model in the similarity measurement using IFC. The paper points out optimization tendencies for the current research, which will be identified and dealt with in the current research. Based on this, the other processes of the CBR move into the focus of the upcoming research.

CBR is based on the fact that there are enough variants in the graph data pool. In the presented example the number of variants is quite limited, but sufficient to show the retrieval process of finding a suitable variant for a simplified given problem. Machine learning can be used in the future research to fill up the variant pool and generate a higher number of possible matches.

Following on this the increase in complexity of buildings and their structures will be addressed in the next phase. Studies indicate that that a high number of nodes, relationships and properties does not cause any difficulties for the graph database and thus the utilisation of the CPU and query speeds are only dependent on the implemented hardware [5]. Therefore, CPU utilisation must be tested under various complex conditions. Complementary Grossniklaus et al. [8] highlight the advantages of a graph database (e.g. Neo4j) in relation to a large amount of data, as it allows high data correlation between entities, an efficient query performance and powerful graph-based algorithm.

The exemplary weights of the entities can be extended to the individual quantities and properties in further studies in order to be able to provide even more targeted results. In addition to that, extensions can be made according to the value, based on other parameters, such as fire protection, u-value or energy efficiency.

In addition to this, the similarity calculation can also be extended to provide more accurate results and needs to be implemented so that an automated similarity matching algorithm can be used. The shown calculation can possibly be supplemented by other established similarity calculations.

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```
MATCH (c:IfcElement)<-[]-(d:IfcElement)<-[]-(e:IfcSpatialStructureElement), (d:IfcElement)<-[]-(f:IfcSpatialStructureElement)
WHERE c.Entity="IfcOpeningElement" AND 0.19<=c.Depth<=0.21 AND 0.8<=c.Height<=0.9 AND 1.8<=c.Width<=2.3
    AND d.Entity="IfcDoor" AND d.`Operation Type`="SINGLE_SWING_RIGHT" AND d.IsExternal=false
    AND e.Entity="IfcSpace" AND e.`Room key Usage type`="1 Living"
    AND f.Entity="IfcSpace" AND f.`Room key Usage type`="2 Office work"
    AND size( (d)-[:IfcRelSpaceBoundary]-() ) >= 2
RETURN c.d.e.f.ID(d)
```

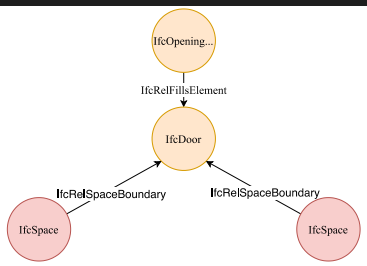


图7. 可选产品变体（PM）

在给定案例中，用户能够基于特定的偏差条件，从基于Neo4j的数据库中检索出一个可选的产品变体（门）。通过执行所提供的Cypher查询，系统会在数据库中搜索与给定墙体洞口尺寸（深度、高度、宽度）相匹配的门。该对象为一樘向右开启的室内平开门，用于连接办公空间与居住空间。系统将在数据库中查找与所选变体相匹配的精确或近似模式，并且该变体具有相应的类型选项。

系统会返回匹配的备选方案，用户可从中选择合适的选项或扩展搜索范围。该流程可根据各产品变体进行调整，并在必要时进一步扩展至功能型和结构型变体。

4 结论与未来工作

借助案例推理技术，可以利用已有知识为类似问题寻找解决方案。相似度度量方法可根据用户的个性化需求灵活调整。

本文提出了一种基于BIM的解决方案，用于建筑早期设计阶段的检索过程。该方案使得能够将存储于BIM模型中的所有信息纳入基于IFC的相似性度量中。同时，本文指出了当前研究中存在的优化方向，这些方向将在后续研究中予以明确并加以解决。在此基础上，案例推理（CBR）的其他环节将成为未来研究的重点。

案例推理方法基于图数据池中存在足够数量的案例这一事实。在所展示的示例中，案例数量虽然较为有限，但已足以说明针对简化给定问题检索合适案例的过程。在未来的研究中，可借助机器学习扩充案例库，并生成更多可能的匹配结果。

在此基础上，建筑物及其结构复杂性的提升将在下一阶段予以探讨。研究表明，大量节点、关系和属性并不会对图数据库的运行造成任何障碍，因此其利用率CPU的性能和查询速度仅取决于所实现的硬件架构[5]。因此，必须在多种复杂条件下对CPU利用率进行测试。此外，Grossniklaus等人[8]强调了图数据库（如Neo4j）在处理海量数据时的优势，因为它能够实现实体间的高度数据关联、高效的查询性能以及强大的基于图的算法支持。

在后续研究中，可将各实体的示例权重扩展至各项具体指标与属性，以获得更为精准的评估结果。此外，还可基于其他参数（如防火性能、传热系数或能源效率等）对其权重进行进一步调整与扩展。

此外，相似度计算还可进一步扩展以获得更为精确的结果，并需加以实现，以便应用自动化的相似度匹配算法。所展示的计算方法亦可辅以其他已确立的相似度计算方法。

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